

Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

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The Problem

Fuel estimation in Philippine agricultural mechanization relies on static coefficients.

- ▶ Generalized coefficients ignore machine variability
- ▶ No localized field, soil, or operator effects
- ▶ Nonlinear interactions invisible to deterministic formulas
- ▶ Result: weak planning, missed efficiency gains

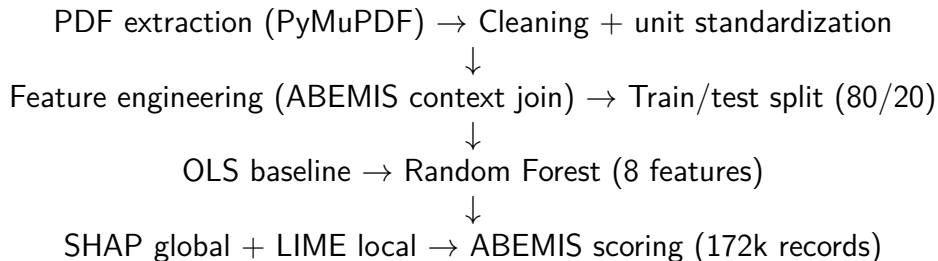
Goal

- ▶ Develop an ML model for fuel-requirement estimation
- ▶ Compare against OLS baseline using R^2 , MAE, RMSE
- ▶ Score national ABEMIS deployments for regional fuel planning
- ▶ Interpret with SHAP (global) and LIME (local)

The Data

- ▶ AMTEC test reports (text + OCR) → **376** cleaned training records
- ▶ ABEMIS inventory → **246,741** fuel-relevant records
- ▶ Coverage: **all 17** Philippine regions
- ▶ **8** RF features: `power_kw`, `year`, three ABEMIS context features, three categorical-encoded (machinery type, family, subset)
- ▶ Target: fuel consumption (L/hr)

Pipeline



Methodology Highlights

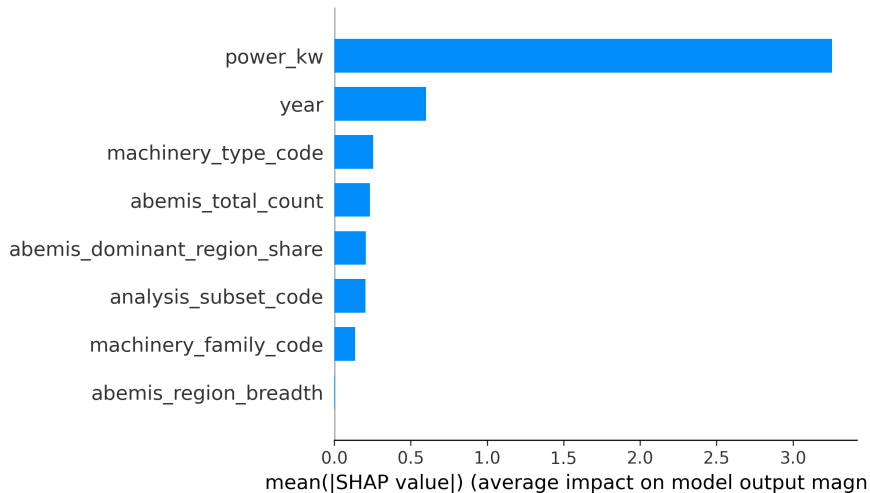
- ▶ **OLS baseline** — linear/quadratic; per-type fallback hierarchy (type → family → global)
- ▶ **Random Forest** — scikit-learn defaults, 80/20 split, 8 features, 5-fold CV
- ▶ **Interpretability** — SHAP TreeExplainer (bar + beeswarm), LIME tabular explainer (10 example records)

Results — Model Comparison ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.704	1.882	2.875
OLS Hierarchical Fallback	0.703	1.879	2.879
Random Forest	0.916	0.950	1.534

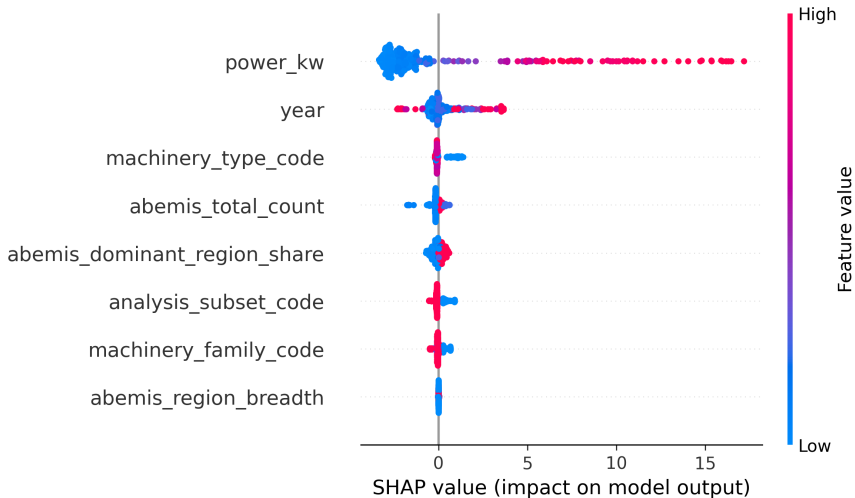
Random Forest captures non-linear power-fuel interactions OLS cannot.

SHAP — Global Feature Importance



power_kw dominates at ~76% importance.

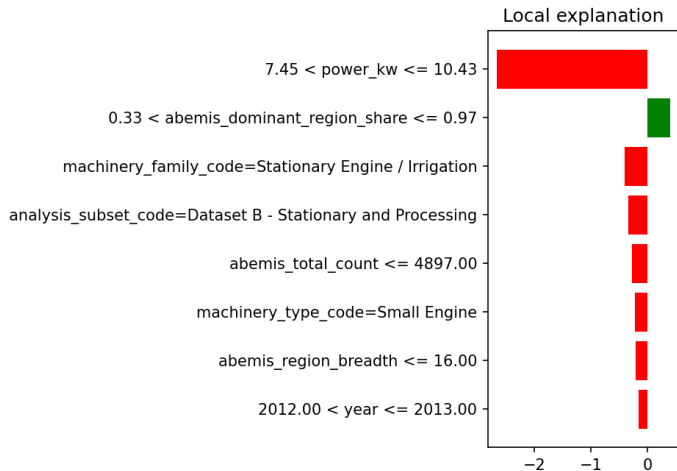
SHAP — Direction & Magnitude



Higher power → higher predicted fuel; context features add nuance.

LIME — A Local Explanation

LIME - Sample 372 | Small Engine | Actual=5.240 Pred=4.773



Per-record explanation builds trust for BAFE engineers.

ABEMIS Regional Scoring

- ▶ **172,265** of 246,741 records successfully scored (**70%** coverage)
- ▶ Excluded: 62,808 missing Rated Power, 11,668 unmappable type
- ▶ Outputs: per-region, per-region $\times$ type, per-record fuel estimates
- ▶ Enables data-driven mechanization planning at national scale

Limitations & Future Work

- ▶ OCR retraining on 247 scanned PDFs (Tesseract binary pending)
- ▶ RAED cropping calendar integration as the Time predictor
- ▶ Expand ABEMIS $\rightarrow$ AMTEC type mapping for $\sim 19,674$ unmapped rows
- ▶ Validation against in-field telemetry from a pilot deployment

Thank you

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